IT303 Software Engineering Course Project

Secure File Sharing with Encryption

User Manual

This guide walks through the actions available on the Secure File Sharing website, including account access, encrypted transfers, key management, account settings, and administrator audit tools. Each step is paired with an annotated screenshot from the site.

Before you begin

Use an email account you can access for verification and password recovery. The site describes file encryption as occurring in the browser before upload. The examples in this guide have been anonymized: private values, usernames, email addresses, keys, and verification codes are hidden.

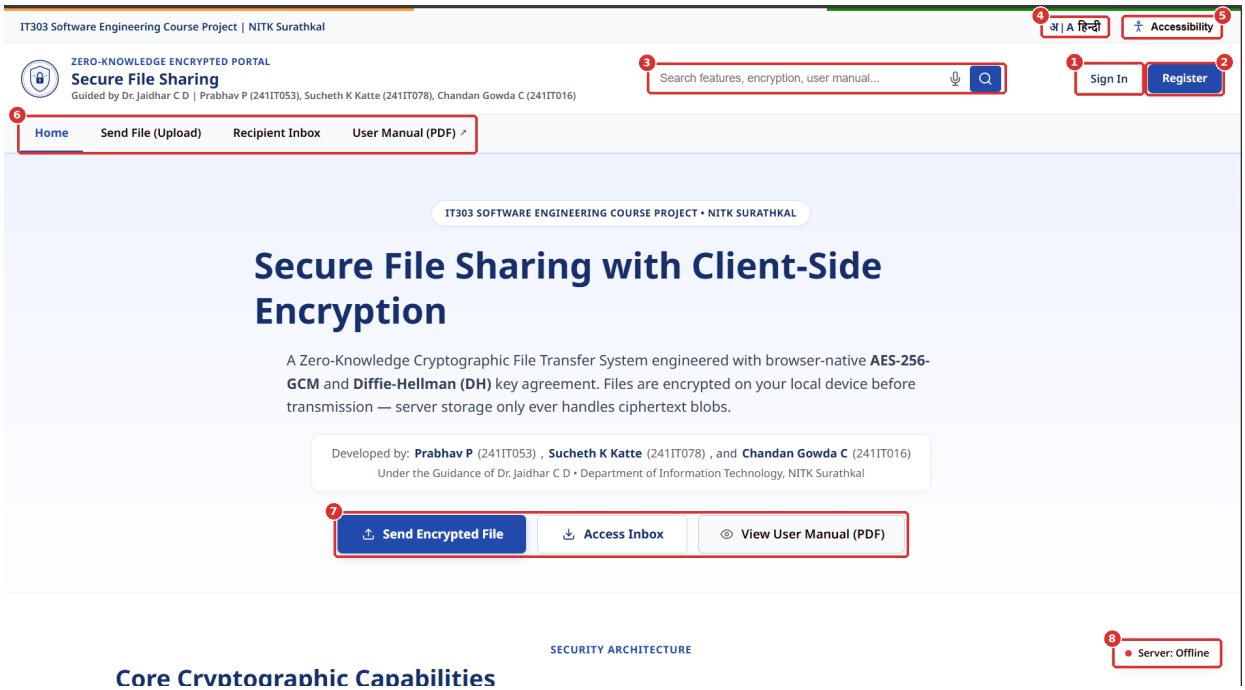
At a glance

Register or sign in; verify your account by email when prompted; send files from Send File (Upload); receive them in Recipient Inbox; back up your private key in the Key Vault; manage credentials in Change Password; and use the Admin Console for account and audit review when your account has the administrator role.

Live site: https://secure-file-sharing-with-encryption-tau.vercel.app/
Prepared: 16 October 2026

1. Start at the home page

Open the live site. The header offers **Sign In** (1) and **Register** (2); the search bar (3), language toggle (4), and **Accessibility** (5) controls are always available. The navigation bar (6) links to Home, Send File (Upload), Recipient Inbox, and the User Manual (PDF). The main buttons (7) start a transfer, open the inbox, or open this manual. The indicator at the lower right (8) shows whether the server is online.

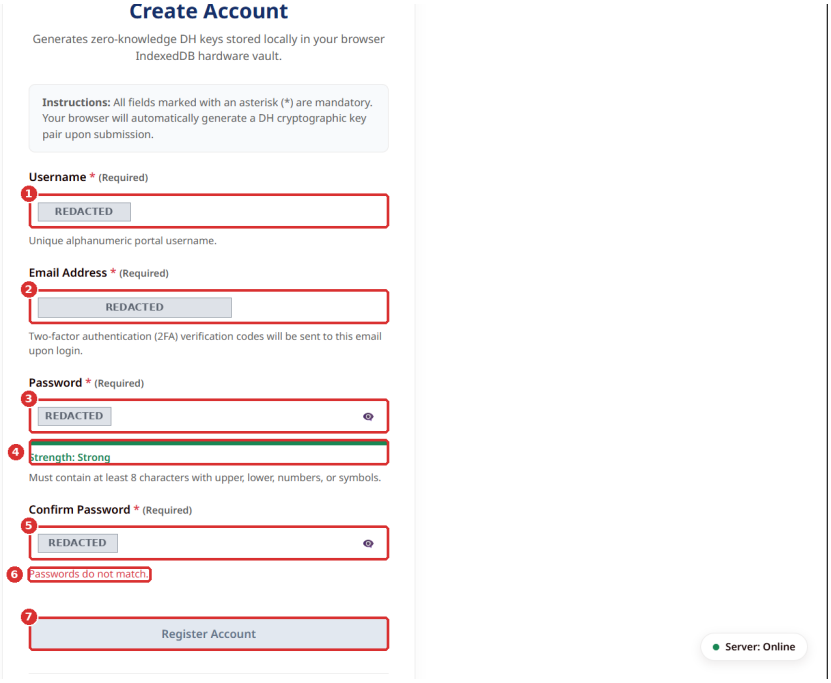


Home page. (1) Sign In, (2) Register, (3) search, (4) language toggle, (5) Accessibility, (6) navigation bar, (7) main action buttons, (8) server status.

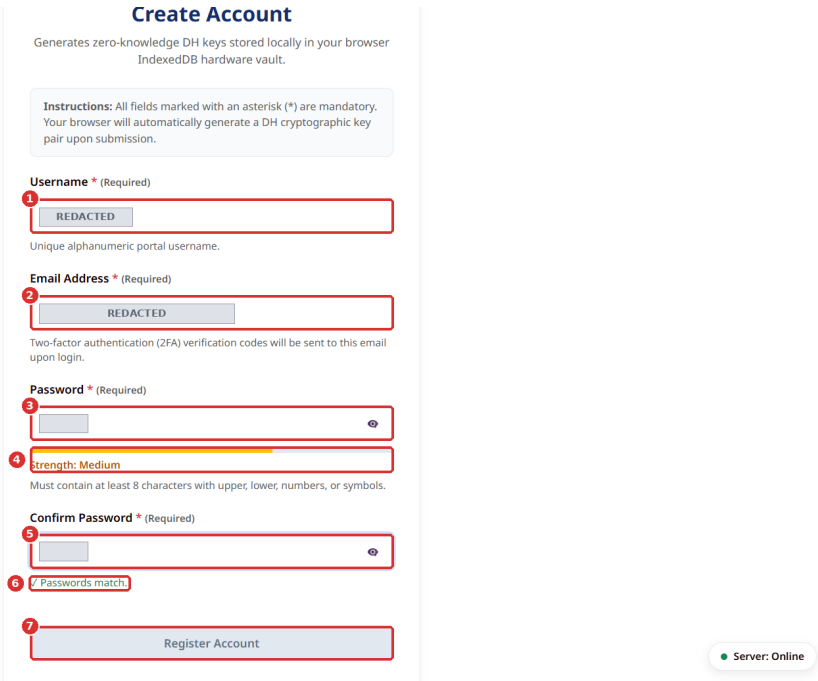
2. Create an account

1. Select **Register** from the home page.
2. Enter a unique alphanumeric **Username** (1) and an **Email Address** (2) you can access. Two-factor verification codes are sent to this address at login.
3. Enter a **Password** (3) of at least 8 characters, mixing upper-case, lower-case, numbers, or symbols. The strength indicator (4) shows the result.
4. Enter the password again in **Confirm Password** (5). The message (6) reports whether the two entries match; the eye icons let you check what you typed.
5. Select **Register Account** (7). The button stays inactive until the form is valid.

Your browser generates a Diffie–Hellman key pair on submission and stores it in the browser's IndexedDB vault. Keep access to the same browser profile and avoid clearing site data unless you have backed up your private key (Section 8).



Registration form with a mismatch warning. (1) Username, (2) email, (3) password, (4) strength indicator, (5) confirmation, (6) match message, (7) Register Account. Values are hidden.



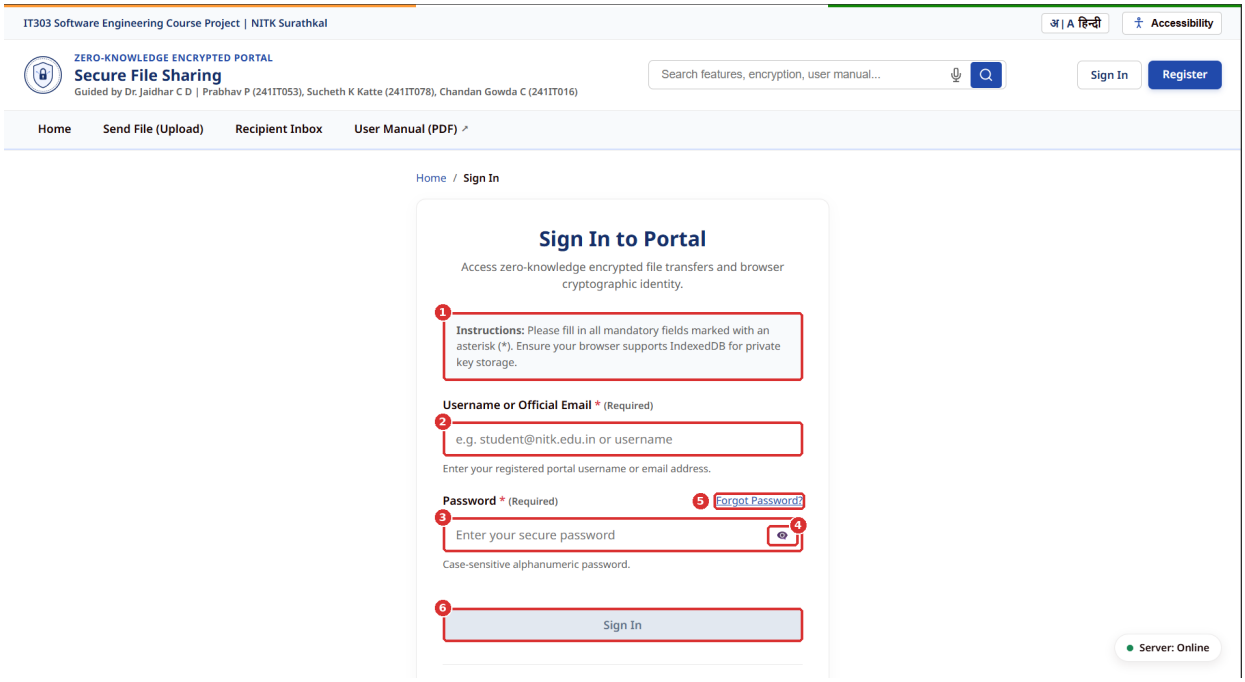
Registration form after the confirmation matches. The same fields are outlined; entered values are hidden.

3. Sign in

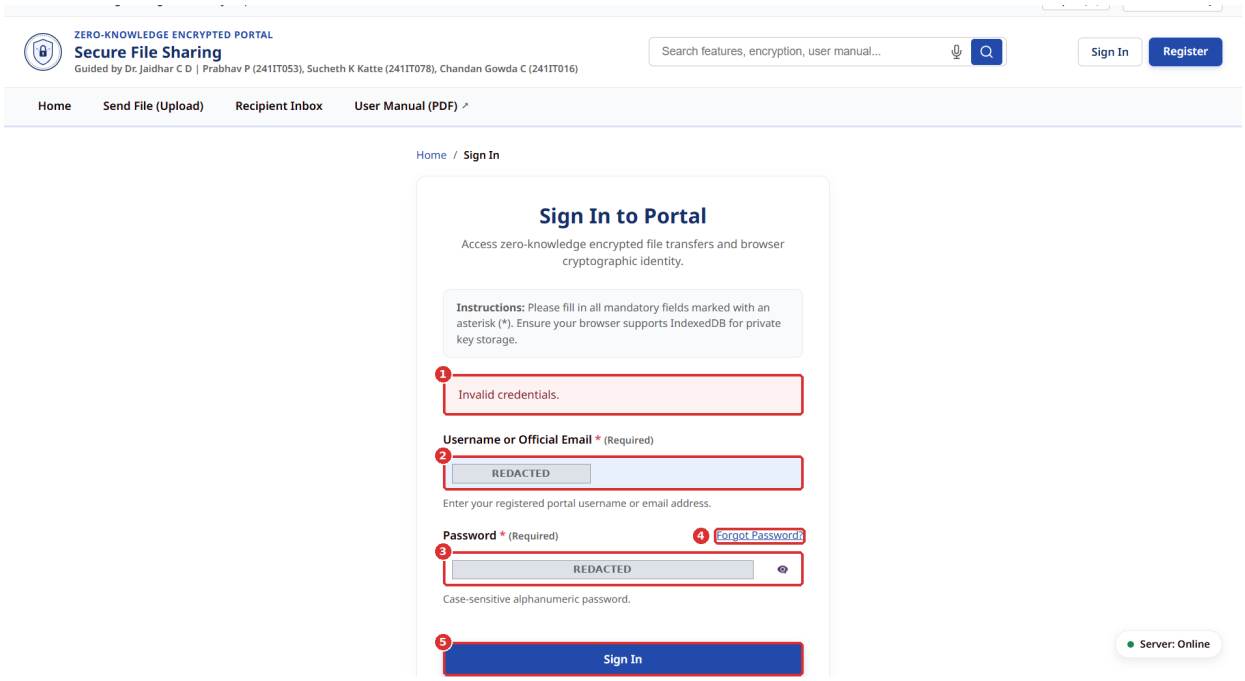
1. Select **Sign In** in the header.
2. Read the instructions panel (1); the page expects a browser that supports IndexedDB for private-key storage.
3. Enter your username or official email (2), then your password (3). The eye icon (4) shows or hides the password while you type.

4. Select **Sign In** (6). If the site requests an extra verification step, continue to Section 4.

If the credentials are wrong, the page shows an **Invalid credentials.** notice (1) above the form; correct the entries and try again. **Forgot Password?** (5) opens password recovery (Section 5).



Empty sign-in form. (1) Instructions, (2) username or email, (3) password, (4) show/hide password, (5) Forgot Password?, (6) Sign In.



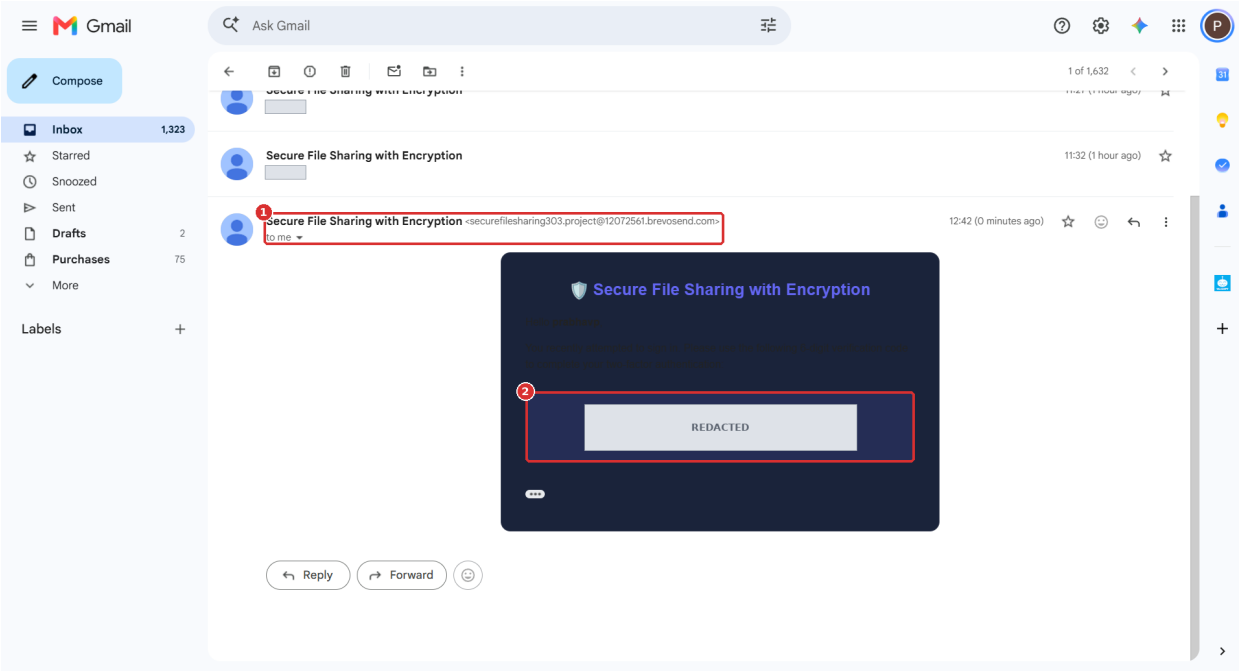
Failed sign-in. (1) Invalid-credentials notice, (2) username field, (3) password field, (4) Forgot Password?, (5) Sign In. Entered values are hidden.

4. Complete email verification

When prompted, the site sends a six-digit verification code to the registered email address.

1. Open the email from **Secure File Sharing with Encryption** (1) and note the current code (2).
2. Enter all six digits in the code field. Use **Paste from Clipboard** if you copied the code.
3. Select **Verify Code & Sign In**. If the code has expired or is missing, use **Resend Code** when it becomes available.

Treat the code as private: do not share it or include it in a message to anyone. Older codes in your inbox no longer apply.



Verification email. (1) Sender and subject line, (2) six-digit code. The code and earlier codes in the inbox list are hidden.

5. Recover a forgotten password

1. On the sign-in form, select **Forgot Password?**.
2. Enter the email address associated with your account and choose **Send Reset Link**.
3. Check your inbox for a message from Secure File Sharing with Encryption. Open it and select **Reset Password**.
4. If you return to the sign-in page, use **Back to Login**.

The site displays a general confirmation that a link was sent if an account exists for the address.

6. Set a new password from a reset link

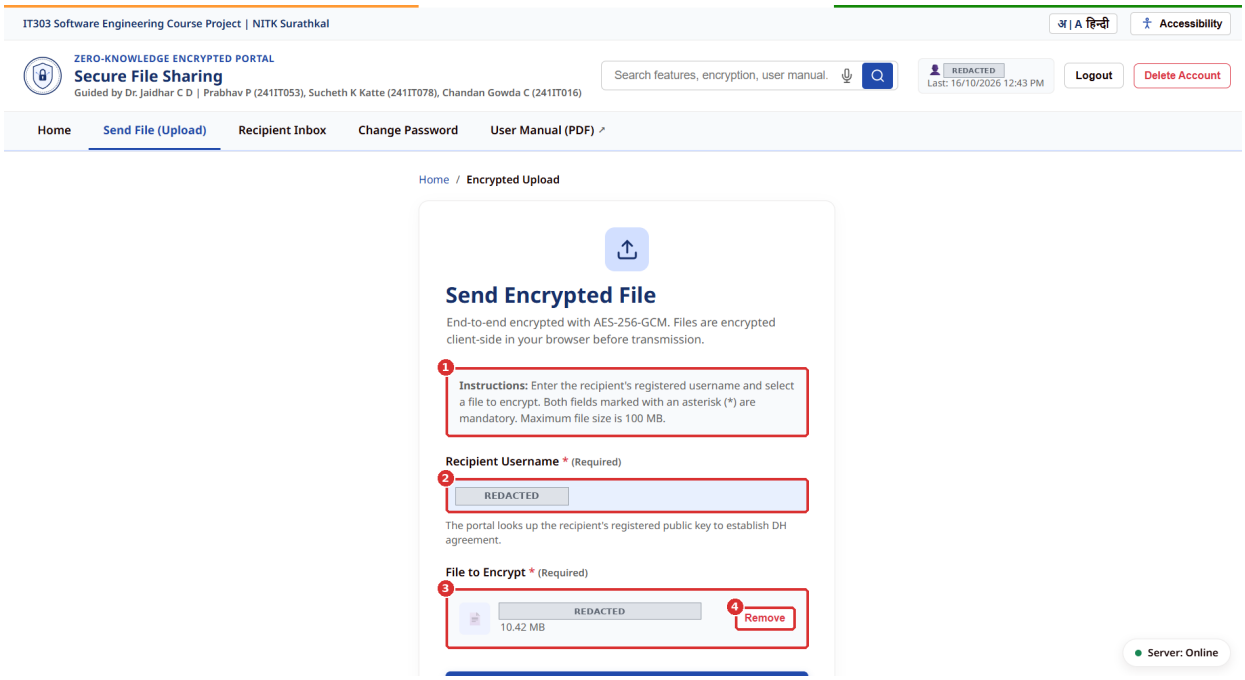
1. Open the reset link from the email before it expires.
2. Enter a new password, then enter the same value in **Confirm Password**.
3. Check the password-strength indicator and correct any warning shown.
4. Select **Reset Password**. If the page reports a mismatch, make both entries identical and try again.

7. Send an encrypted file

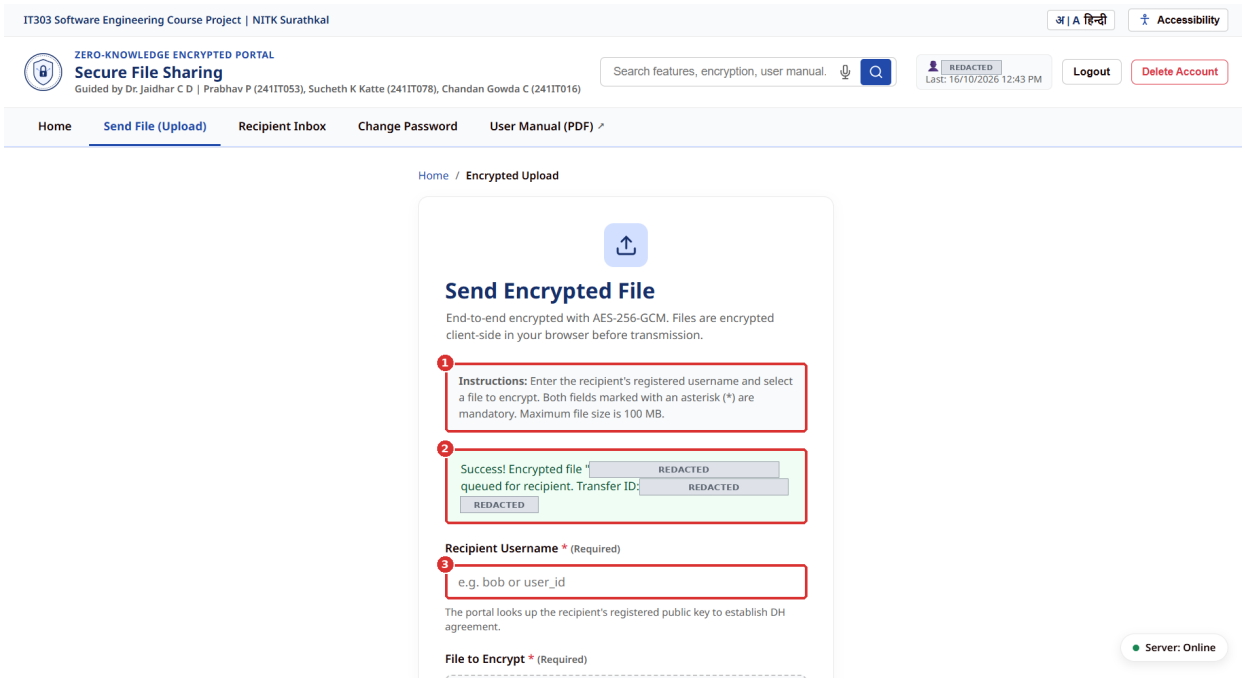
Select **Send File (Upload)** from the signed-in navigation bar. Files are encrypted with AES-256-GCM in your browser before transmission; the maximum file size is 100 MB.

1. Read the instructions **(1)**. Both fields marked with an asterisk are mandatory.
2. Enter the recipient's registered **username** **(2)**. The portal looks up the recipient's registered public key to establish the Diffie–Hellman agreement. Check the spelling with the recipient.
3. Select the file to encrypt, or drag and drop it. The chosen file and its size appear in a row **(3)**; use **Remove** **(4)** to pick a different file.
4. Confirm the file and recipient, then select **Encrypt & Queue Transfer**.
5. Wait for the green success notice **(2)**, which names the file and gives a transfer ID. The recipient can then check their Inbox; the form resets for the next transfer **(3)**.

Send only to the intended username. Screenshots conceal the example recipient, filename, and transfer identifier.



Upload form ready to send. (1) Instructions, (2) recipient username, (3) chosen-file row, (4) Remove. Personal values are hidden.



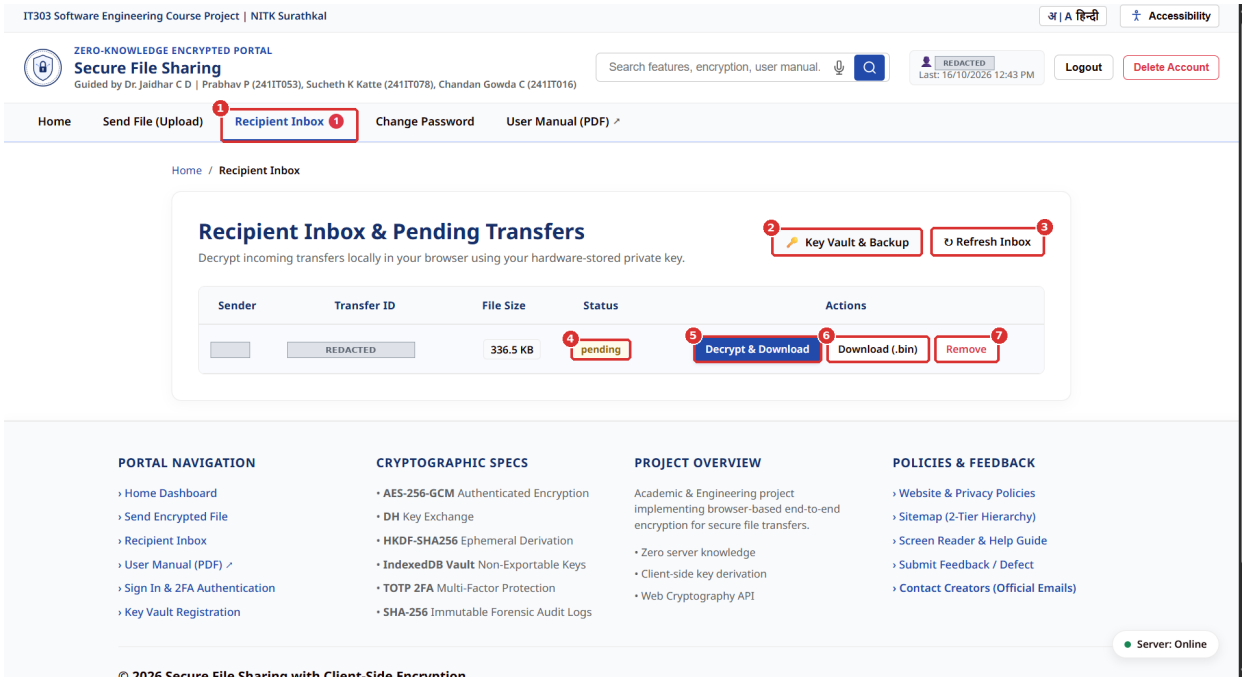
After a successful queue action. (1) Instructions, (2) success notice, (3) cleared recipient field. File name and transfer ID are hidden.

8. Receive a file

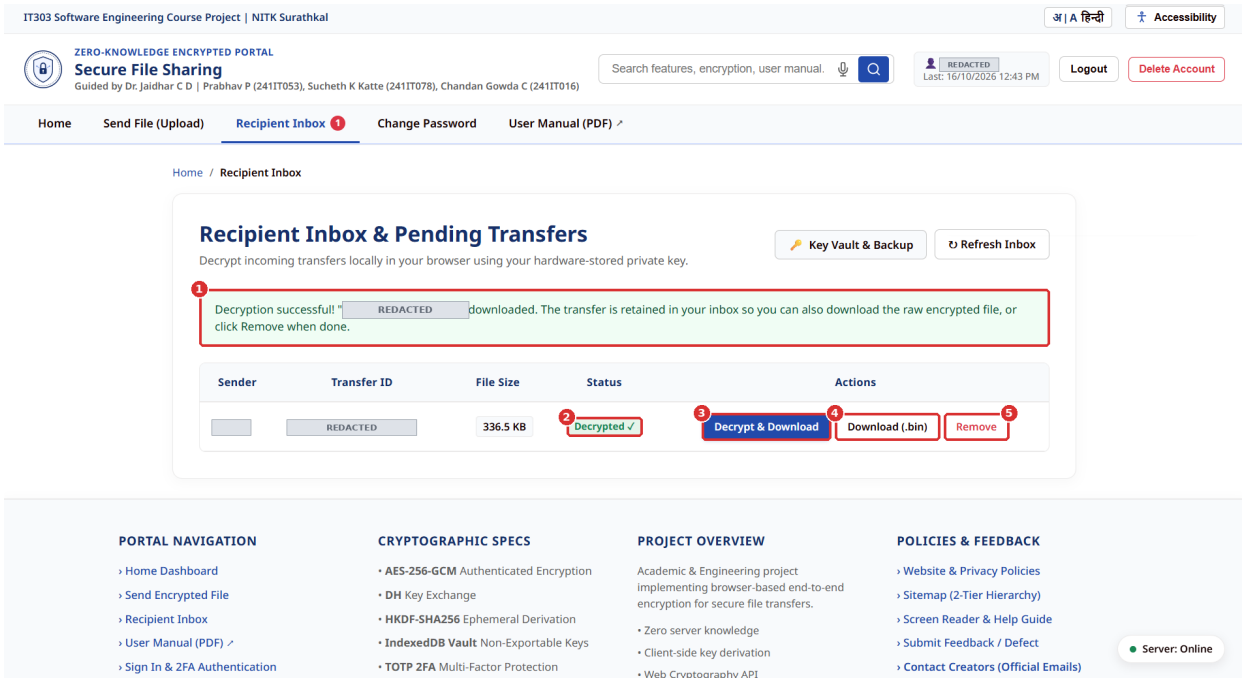
Select **Recipient Inbox** (1); a red badge shows how many transfers are waiting. Choose **Refresh Inbox** (3) if an expected transfer is not listed. Each row lists the sender, transfer ID, file size, and status (4).

1. Check the sender and transfer with the person who sent the file.
2. Choose **Decrypt & Download** (5) to decrypt the file in your browser with your private key and save the readable file.
3. Or choose **Download (.bin)** (6) to save the raw ciphertext.
4. When you no longer need the transfer, select **Remove** (7). The inbox confirms that the transfer was removed and the server copy permanently purged.

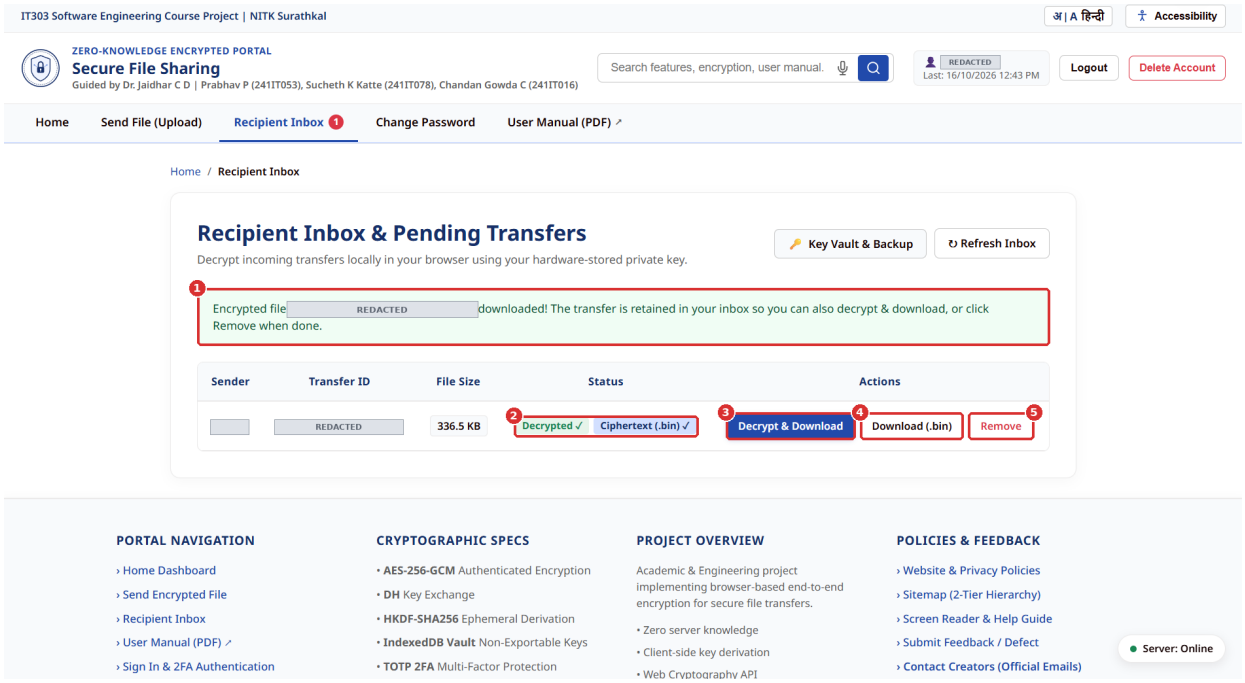
A transfer stays in the inbox after you download it, so you can decrypt it again or download the ciphertext, until you remove it. The status column changes from **pending** to **Decrypted** and **Ciphertext (.bin)** as you act on the transfer.



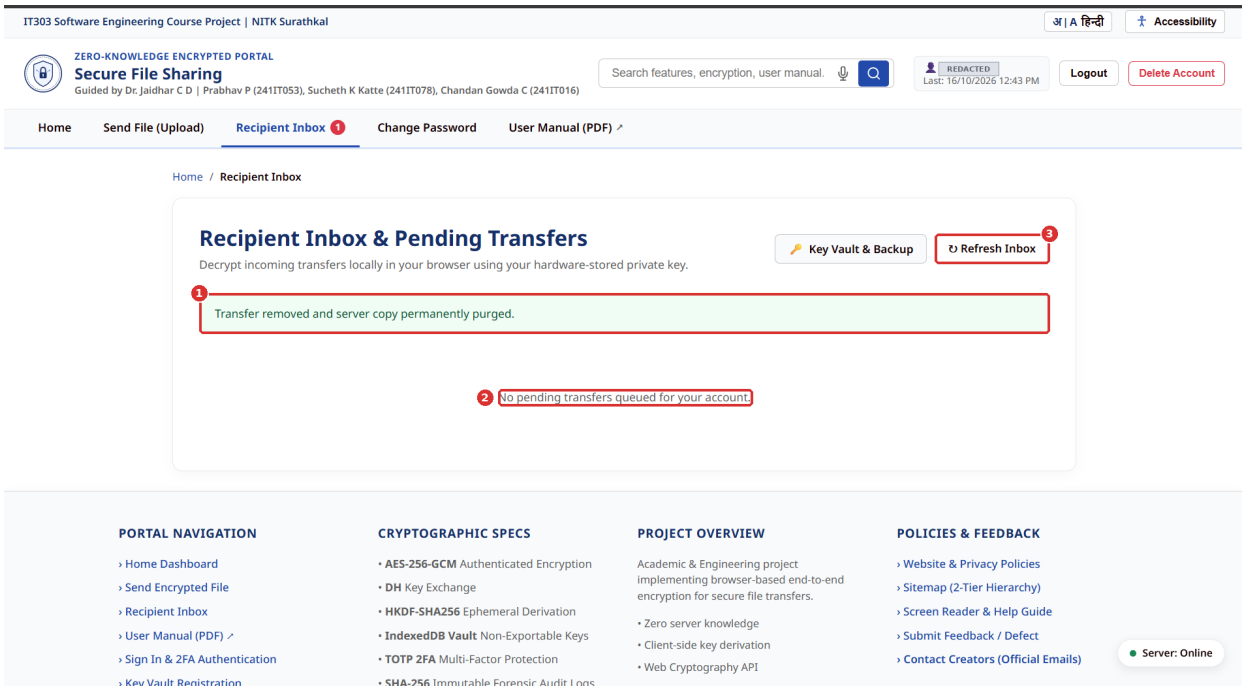
Pending transfer. (1) Recipient Inbox tab, (2) Key Vault & Backup, (3) Refresh Inbox, (4) status, (5) Decrypt & Download, (6) Download (.bin), (7) Remove. Sender and transfer ID are hidden.



After decrypting. (1) Success notice, (2) Decrypted status, (3) Decrypt & Download, (4) Download (.bin), (5) Remove. File name, sender, and transfer ID are hidden.



After also downloading the ciphertext. (1) Notice, (2) Decrypted and Ciphertext (.bin) status marks, (3–5) the same three actions. Names and IDs are hidden.



After removing the transfer. (1) Purge confirmation, (2) empty-inbox message, (3) Refresh Inbox.

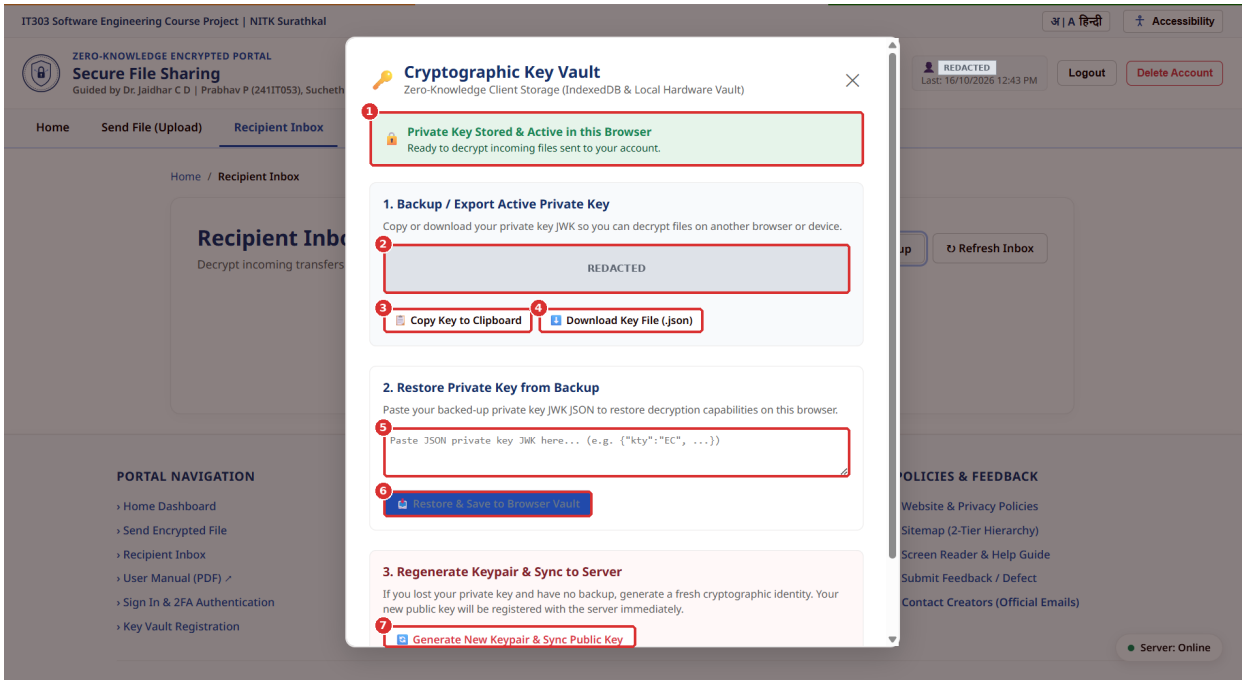
9. Back up and restore your private key

Select **Key Vault & Backup** in the inbox. The vault panel confirms when a private key is stored and active in this browser (1).

1. **Back up:** copy the private key JWK (2) with **Copy Key to Clipboard** (3), or save it with **Download Key File (.json)** (4). Store it somewhere safe and private; anyone holding it can decrypt files sent to you.

2. **Restore:** on another browser or device, paste the backed-up JWK into the restore box (5) and select **Restore & Save to Browser Vault** (6).

3. **Regenerate:** if you lost your private key and have no backup, **Generate New Keypair & Sync Public Key** (7) creates a fresh identity and registers the new public key with the server. Files encrypted to the old key cannot be decrypted with the new one.

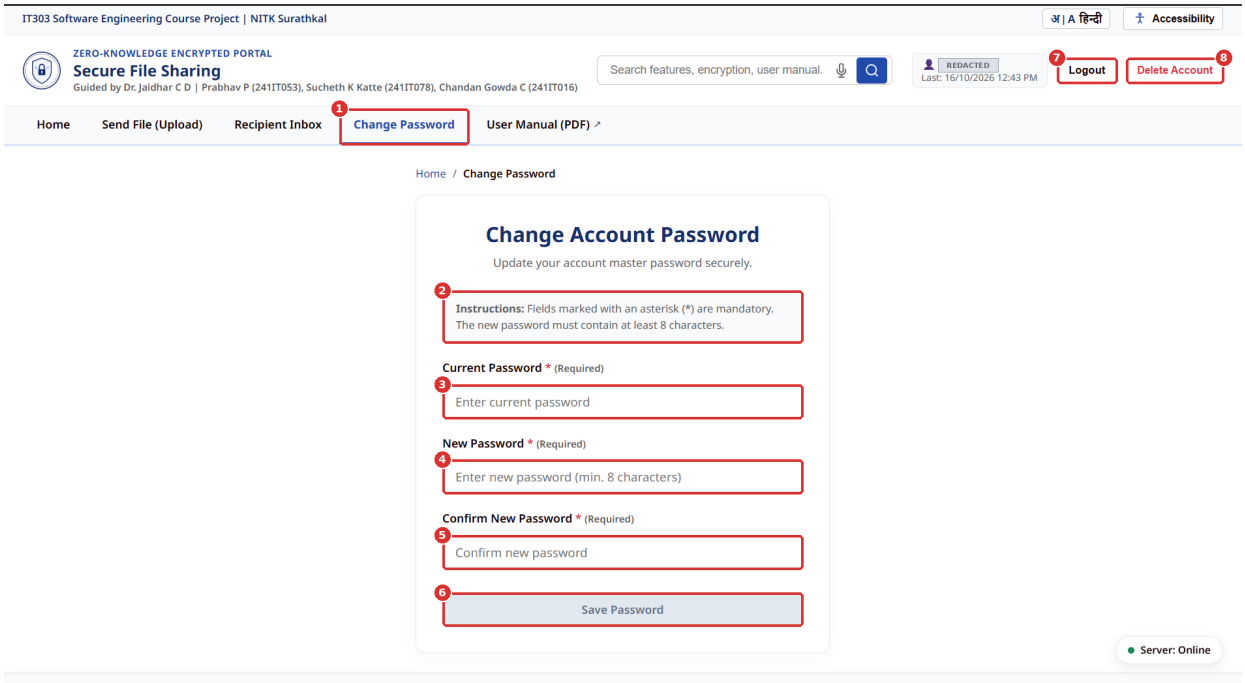


Cryptographic Key Vault. (1) Status, (2) exported key, (3) Copy, (4) Download, (5) restore box, (6) Restore, (7) Regenerate. The key itself is hidden.

10. Change your password

Select **Change Password** (1) in the navigation bar. Fields marked with an asterisk are mandatory and the new password needs at least 8 characters (2).

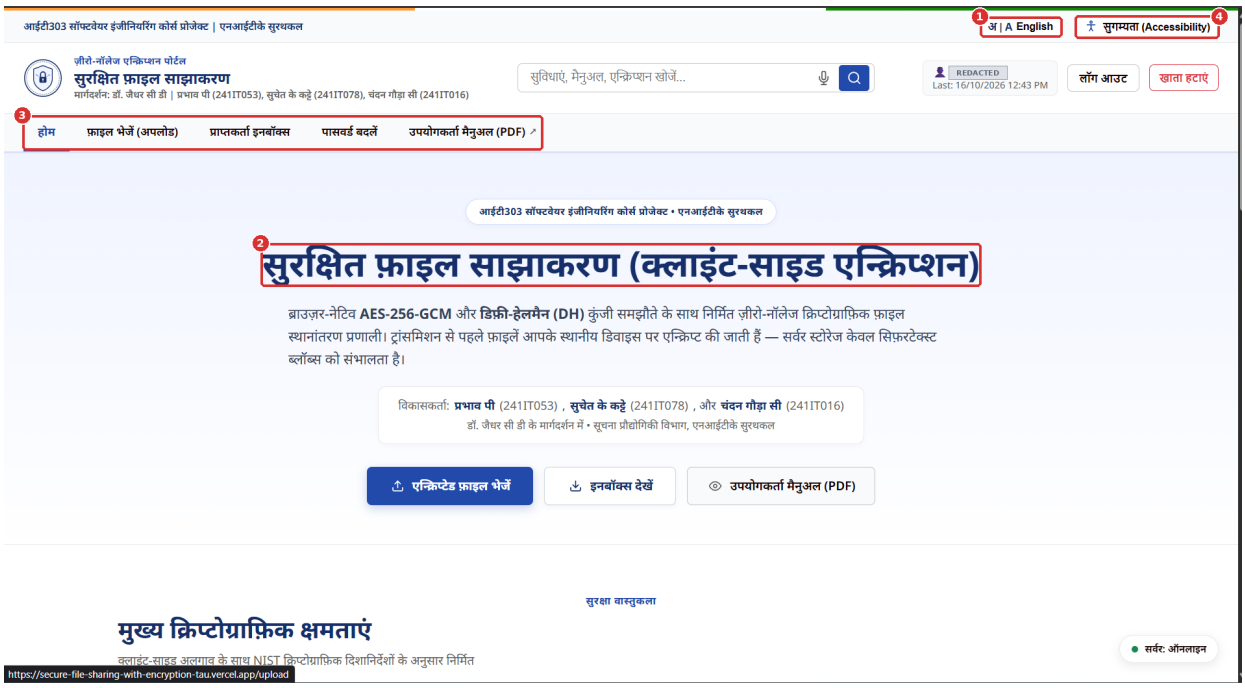
1. Enter your current password (3).
2. Enter a new password (4) and confirm it (5).
3. Select **Save Password** (6).



Change-password page. (1) Tab, (2) instructions, (3) current, (4) new, (5) confirm, (6) Save Password, (7) Logout, (8) Delete Account. Account name is hidden.

11. Change language and use search and accessibility tools

Language. The toggle at the top right (1) switches the whole site between English and Hindi, including the headings (2) and navigation (3).



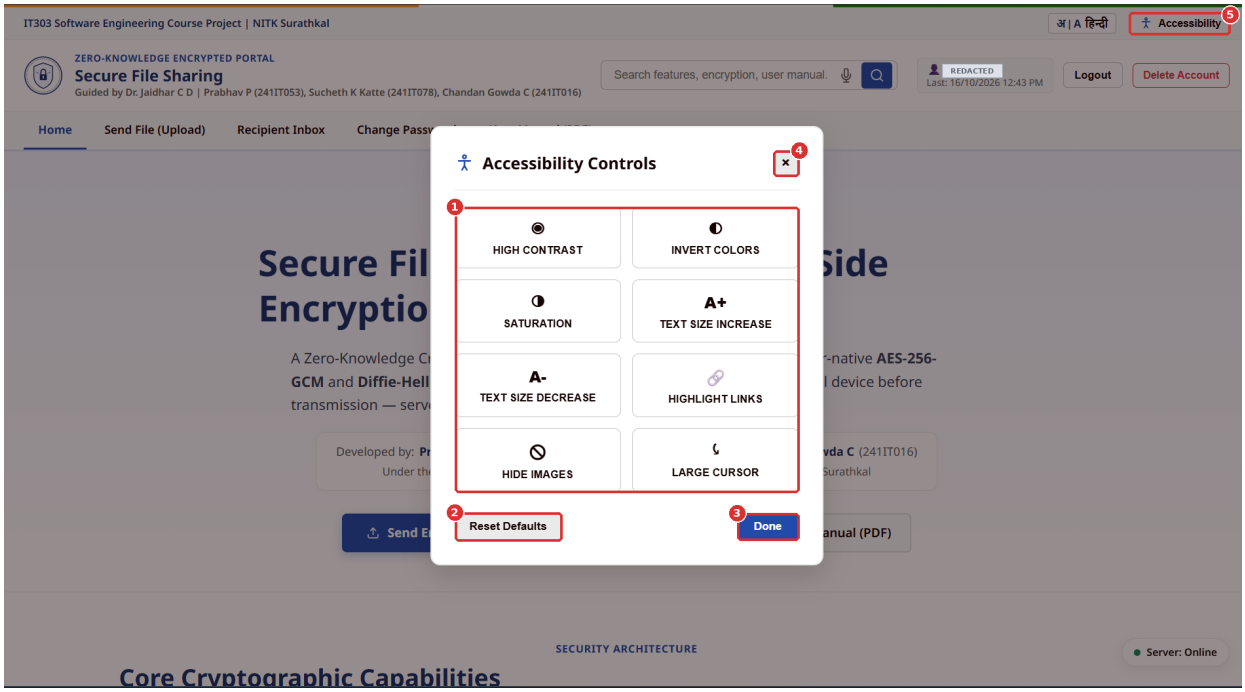
Home page in Hindi. (1) Language toggle, (2) translated heading, (3) translated navigation, (4) Accessibility.

Search. Select the header search box to open the search panel. Type in the field (1) or browse the category tabs (3) (All, Services, Documentation, Security, Account); the results list (4) links to pages and manual topics and the footer (5) shows the match count. Press **ESC** (2) to close it.



Search panel. (1) Search field, (2) ESC, (3) categories, (4) results, (5) result count.

Accessibility. The **Accessibility** button (5) opens a panel of display aids (1): high contrast, invert colors, saturation, text size increase and decrease, highlight links, hide images, and large cursor. Use **Reset Defaults** (2) to undo changes and **Done** (3) or the close button (4) to dismiss the panel.



Accessibility controls. (1) Option grid, (2) Reset Defaults, (3) Done, (4) close, (5) Accessibility button.

12. Log out

Select **Logout** (7) in the header, beside your account name (see the change-password screenshot). You return to the public part of the site; sign in again when you want to continue.

13. Delete your account

Account deletion is permanent. The confirmation warns that all keys and pending transfers will be purged.

1. Select **Delete Account** (4) only when you intend to remove the account.
2. Read the browser confirmation (1) carefully. Choose **Cancel** (3) to keep the account, or **OK** (2) only if you accept the deletion.
3. After confirming, the site returns to the public home page.



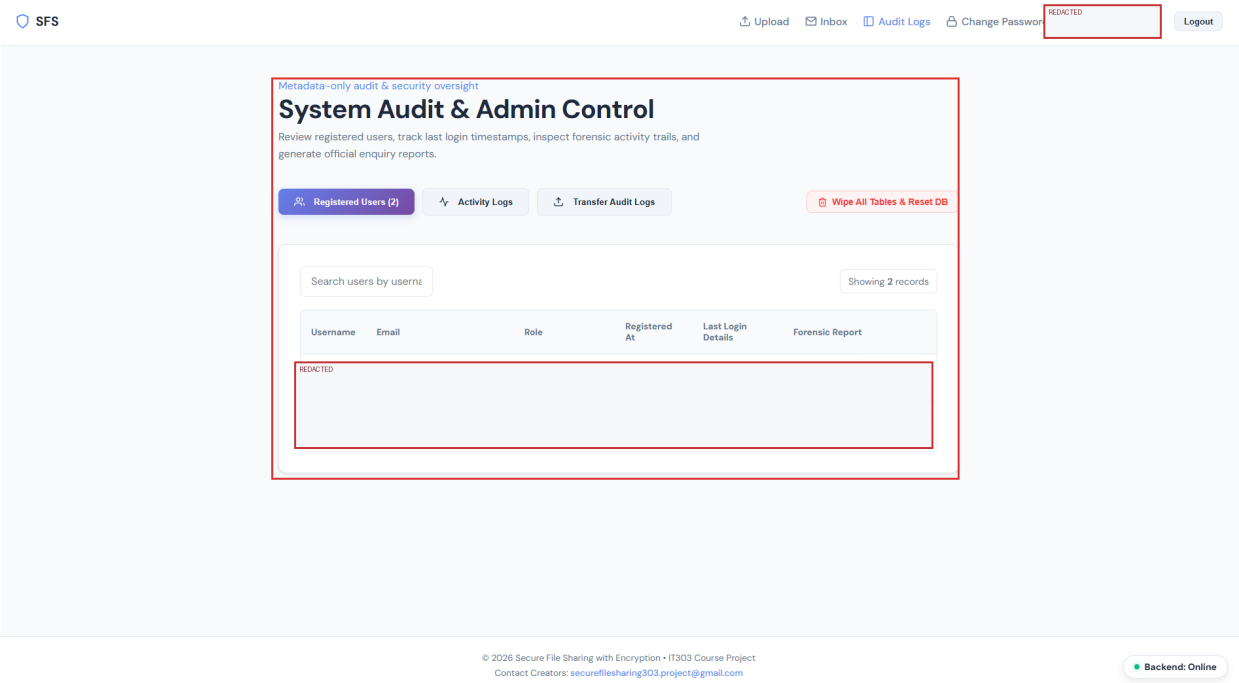
Deletion confirmation. (1) Warning dialog, (2) OK, (3) Cancel, (4) Delete Account. Review the warning and choose deliberately.

14. Administrator: review users and audit activity

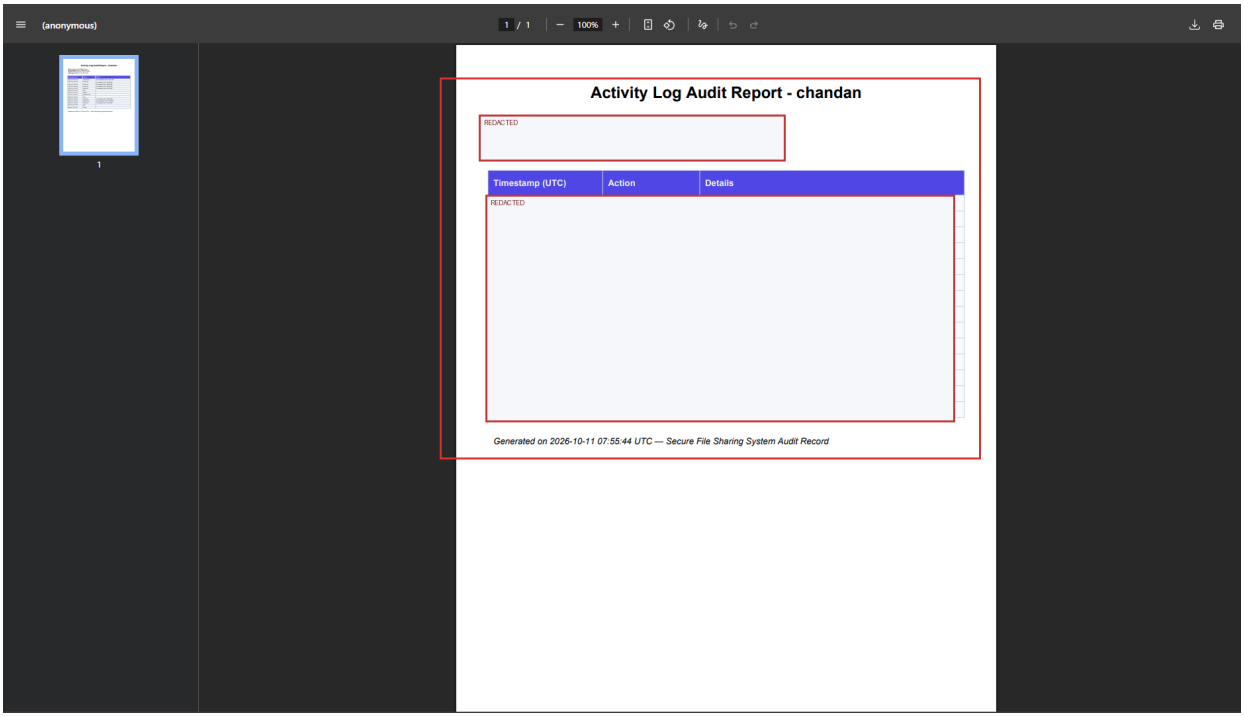
These actions are available only to an administrator. Open **Admin Console** (1) in the navigation bar to reach **System Audit & Security Administration**, a metadata-only oversight page.

1. Use the **Registered Users** tab (2) to list accounts, with username, email, role, registration date, and last login. Search by username in the box (6).
2. Select **View Logs** (7) to inspect a user's activity, or **Export PDF** (8) to produce an audit report for that user.
3. Use the **Activity Logs** (3) and **Transfer Audit Logs** (4) tabs to review the corresponding records.

The exported report lists the account's email, status, registration and last-login times, and a table of timestamped actions such as register, login, and upload.



Administrator dashboard. (1) Admin Console, (2) Registered Users, (3) Activity Logs, (4) Transfer Audit Logs, (5) Reset Database Tables, (6) search, (7) View Logs, (8) Export PDF. Usernames and emails are hidden.



Exported audit report. (1) Title, (2) account summary, (3) action table, (4) generation footer. User details are hidden.

15. Administrator: reset the database

The dashboard's **Reset Database Tables** button (5) (previous screenshot) is destructive. Its confirmation warns that user accounts, transfers, logs, and stored encrypted blobs will be permanently removed. Read the complete confirmation before proceeding. Cancel if you are not authorized to reset the database or do not intend to erase this data.

Screenshot key

Red rectangles with numbered circles identify the controls discussed in the nearby steps; the numbers match the **(n)** markers in the text and the caption lists. Pale grey panels marked **REDACTED** conceal private values from the supplied examples. Screens show example states; available records, messages, and controls depend on the account and current site state.